

Week 2 – Workshop

Physical Properties, Mixing Laws, and Asking the Right Questions



Geophysics 3A: Potential Fields & Geothermics

July 10, 2026

Duration: 3 hours

Setting: Workshop room

Preparation

Pre-reading: Course Notes Ch. 2

Lecture videos:

Notes:

Purpose

This workshop develops intuition for how physical properties (density ρ , thermal conductivity k , thermal diffusivity κ , bulk modulus K_T , shear modulus μ , seismic velocities V_P , V_S , magnetic susceptibility χ , and heat production A) depend on state variables (pressure P , temperature T , composition, porosity/cracks), and how geometry and fabric determine appropriate mixing relations (arithmetic, geometric, harmonic; Voigt–Reuss–Hill). Students practice cross-disciplinary collaboration and formulate the key questions a practicing geophysicist must ask a geologist before modeling or surveying.

Learning Outcomes

- Select and justify mixing laws based on rock fabric (random granular, layered/foliated, fractured) and target property (bulk vs. transport vs. elastic).
- Reason about how P – T –composition/porosity shift properties and create anisotropy.
- Identify which properties most strongly influence gravity, magnetic, and heat flow interpretations.
- Elicit geological information via well-posed questions to constrain non-uniqueness.

Session Flow (170 min)

Create mixed groups (4–5 students each) with both geology and physics/engineering backgrounds.

0–10 min: Warm-up — What does geophysics see?

Much like our senses, different fields are sensitive to different things. Are geophysical fields sensitive to individual minerals or bulk rock properties? Are they sensitive to different scales?

10–30 min: Mini-lecture and Q&A

- Bulk-additive vs. pathway-limited vs. elastic-interaction properties.
- Mixing relations: arithmetic (parallel), harmonic (series), geometric (random granular), Voigt–Reuss–Hill (elastic bounds).
- Scale-dependent anisotropy (micro/meso/macro fabrics).

30–45 min: Live Demonstration — “Crowd Conductivity”

Aim: Intuitively illustrate why different mixing laws arise and how anisotropy appears.

1. **Random scattering** (students distributed irregularly): volunteer transits across room. Multiple semi-obstructed pathways $\Rightarrow$ geometric-like effective behavior.
2. **Parallel corridors** (lanes aligned with motion): fastest transit $\Rightarrow$ arithmetic mean (parallel pathways).
3. **Series barriers** (rows perpendicular to motion): slowest transit $\Rightarrow$ harmonic mean (series resistances).
4. **Anisotropy:** compare transit parallel vs. perpendicular to fabric.

Optional: time a handful of transits and note distribution shape differences.

45–60 min: Simulation Walkthrough

Builds on crowd conductivity with computation. We'll examine a pre-built Python visualization of transit times through three media (random, parallel, series) and the resulting histogram shapes. Relate to effective properties and mixing choices.

60–80 min: Activity 2.A: Rock Fabric and Mixing Laws

Distribute a set of photos (thin section, hand sample/outcrop close-up, outcrop/aerial). For each image, groups identify: geometry (random granular/layered/foliated/fractured), appropriate mixing law for density, and conductivity, whether the material is isotropic at that scale, and how conclusions change with scale.

80–125 min: Activity 2.B: When to Estimate Physical Properties

Each group tackles two scenarios (20 min each: 12 min discussion, 5 min notes, 3 min question drafting). Use the dedicated scenario handouts. Outcomes must include: (i) property/mixing-law decisions and (ii) 2–3 *questions to ask the geologist* before committing to a model/survey.

125–155 min: Activity 2.C: Path Simulation

In this computer demonstration, we'll explore how the arrangement of properties affects the path and rate of travel across a medium. While not a true simulator of physical properties, the results will show how various geometries of multicomponent systems affect the path and travel-time.

155–165 min: Wrap-up

What does geophysics see? Geophysics responds to bulk physical properties, not individual minerals. The appropriate mixing law depends on the rock fabric and the property being estimated. Can you name a physical property for each of the geophysical methods introduced so far (gravity, magnetism, heat flow)?

165–175 min: Preview: Inversion

In geophysics, we typically measure fields and infer the underlying physical properties. In this course, however, we mostly focus on the *forward problem*: computing fields from a known distribution of properties.

In practice, we usually face the opposite challenge: the *inverse problem*. Inverse problems are difficult for several reasons:

- they are often *underdetermined*, with fewer observations than unknowns;
- *non-unique*, meaning different models can produce the same data; and
- *unstable*, where small changes in the model can produce large changes in the predicted field.

In addition, many geophysical relationships are non-linear, which adds further complexity.

Next week, we'll begin looking at how inversion methods address these challenges.

Workshop Notes

Workshop Notes

Activity 2.A

Rock Fabric & Mixing Laws

Purpose

This activity develops visual intuition for how rock texture and fabric determine which mixing law is appropriate for estimating bulk physical properties. By classifying a series of rock photographs at different scales, you will connect the geometry of a material to the mathematical model best suited to describe its effective properties.

Learning Outcomes

By the end of this activity, you should be able to:

- Identify the fabric or geometry of a rock from a photograph and classify it as random granular, layered parallel, or layered in series.
- Select the appropriate mixing law (arithmetic, geometric, or harmonic) for density and thermal conductivity given a rock's geometry.
- Determine whether a material is isotropic or anisotropic at a given scale of observation.

Background

Mixing relationships are mathematical methods that we use to approximate the physical behavior of an rock from its constituent mineralogy and porosity. These relationships can sometimes be derived from first principals—as we'll show for harmonic means and thermal conductivity later in the course—though more often they are empirical approximations that capture the messiness of the world.

Physical properties can be divided into two broad classes.

Bulk properties depend primarily on the amount of each constituent material present and are relatively insensitive to how those materials are arranged. Examples include density (ρ) and specific heat capacity (C_P). These properties are generally estimated using volume-weighted arithmetic averages.

Geometry-controlled properties depend not only on the constituent materials but also on how they are distributed and connected. Examples include thermal conductivity (k), electrical conductivity (σ), and many elastic properties (bulk and shear moduli). For these properties, the appropriate mixing relationship depends on the geometry and orientation of the materials.

Table 1: Guidelines for selecting mixing relationships.

Geometry / Fabric	Density	Conductivity	Elastic Moduli
Random granular mixture	Arithmetic	Geometric	Voigt–Reuss–Hill
Layered, parallel to transport	Arithmetic	Arithmetic	Direction-dependent
Layered, perpendicular to transport	Arithmetic	Harmonic	Direction-dependent
Strongly fractured or porous material	Arithmetic	Depends on connectivity; often harmonic-like	Depends on fracture geometry and filling

Instructions:

Open the slide deck (isotropy_anisotropy.ppt) on MyLearning course website. The pictures of rocks range from micrographs to aerial photos. For each photo, ask:

If this were a gravity survey, would the distinction between individual minerals matter?

If this were a conductivity problem, would connectivity matter?

Consider how your interpretation might **change at a different scale**.

For each photo fill in your responses that:

- Identify the **fabric/geometry**.
- Identify the **approximate scale** of the photo.
- Select an appropriate **mixing law for density** (ρ).
- Select an appropriate **mixing law for thermal conductivity** (k).
- State whether the material is **isotropic or anisotropic** at this scale.

Photo	Description	Scale Micro / Macro / Meso	Mixing Law (A)rithmetic, (G)eometric, (H)armonic		(I)sotropic / (A)nisotropic
			Density	Thermal Conductivity	
Example	uniform porous sandstone	Meso	A	G	I
#1					
#2					
#3					
#4					
#5					
#6					
#7					
#8					
#9					
#10					
#11					
#12					
#13					
#14					
#15					

Key Ideas

For bulk rock properties, an arithmetic mean is always used to compute the effective property estimate.

For pathway sensitive properties,

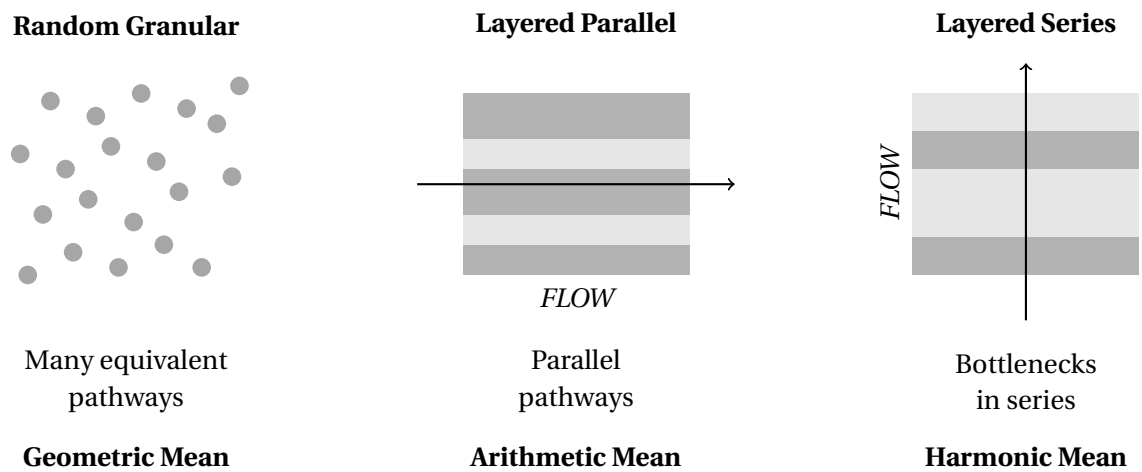


Figure 1: Conceptual relationship between fabric, transport geometry, and appropriate mixing relationship.

Activity 2.B

When to Estimate Physical Properties

Purpose

This activity develops professional judgement in selecting physical property estimation methods for realistic geoscience problems. Through a series of scenarios drawn from basin analysis, tectonics, geodynamics, mineral exploration, and geothermal studies, you will practice identifying the most appropriate method given the available data, the target property, and the geological context.

Learning Outcomes

By the end of this activity, you should be able to:

- Compare the strengths and limitations of different approaches to estimating physical properties (direct measurement, lab data, composition-based estimates, proxies/correlations, geophysical inversion).
- Select and justify an estimation method appropriate to the available data, the scale of the problem, and the target property.
- Formulate targeted geological questions that would reduce uncertainty in a physical property model.
- Identify sources of non-uniqueness and uncertainty in property estimation, including pressure–temperature effects, alteration, and limited sample coverage.

Instructions

Below you will be provided with a set of scenarios that you as a geophysicist are required to develop a physical property model for. The point of this exercise is to develop a model that can be used as a reference based on the available information. In some cases the best and only option may be to run a geophysical survey and model the property from the results, though generally we can use existing samples and datasets to produce an initial property model.

The responses to these scenarios closely align with the lecture notes on methods used to determine/estimate physical properties.

For the discussions, students should be in groups mixed with geologists and physicists. For each scenario:

- draft 2–3 questions that a practicing geophysicist should ask the geologist for additional information about the geologic setting before devising your scenario,
- state your *preferred method* to obtain the required property (e.g., direct measurement, drill-core lab data, composition-based estimate, proxy/correlation, thermodynamic modeling, geophysical inversion/velocity–density regressions), and
- justify why you chose it over other options, list pros and cons, and write 2–3 key *questions to ask the geologist*.

Tip: articulate data availability, scale, P–T effects, alteration/weathering, and non-uniqueness when comparing methods.

Scenario A: Basin Density Model (Resources/Tectonics)

Motivation/Goal. You are supporting a basin analysis that informs *water resource assessment* and subsidence related to extraction. Drill cores are available at multiple wells, and a seismic stratigraphic framework defines a layered structure across the basin. The task is to build a depth-dependent **density model** suitable for forward gravity modeling and basin compaction studies. How will you use the available samples and geophysical data to develop the density model?

Your plan**2–3 Questions for the Geologist****Preferred method and why****Pros**

- _____
- _____

Cons

- _____
- _____

Scenario B: Mountain-Belt Cross-Section (Tectonics)

Motivation/Goal. You are preparing a balanced structural cross-section across a metamorphic belt to support a tectonic reconstruction a for a large civil engineering project (construction of a hydroelectric dam). Rock types are mapped at the surface; access to samples are sparse and uneven, though the region has been explored for mineral resources previously. You need **density** for major units to test isostatic plausibility and gravity fit. How do you produce a density model?

Your plan

2–3 Questions for the Geologist

Preferred method and why

Pros

- _____
- _____

Cons

- _____
- _____

Scenario C: Upper-Mantle Density for Isostasy (Geodynamics)

Motivation/Goal. You are part of a team trying to better understand subduction dynamics. A 3-D regional tomography experiment images V_s and V_p within the mantle down to a depth of 410 km, capturing the mantle wedge, subducted plate and remaining upper mantle. The geodynamic model requires an upper-mantle **density** model as an input. What would be your approach to producing this density model?

Your plan

2–3 Questions for the Geologist

Preferred method and why

Pros

- _____
- _____

Cons

- _____
- _____

Scenario D: Aeromagnetic Susceptibility Model (Mineral/Exploration)

Motivation/Goal. For a greenfields **mineral exploration** program, you have a regional aeromagnetic survey. Your brief is to produce a **susceptibility model** that distinguishes key prospective units and to flag where remanent magnetization may invalidate induced-only assumptions.

Your plan**2–3 Questions for the Geologist****Preferred method and why****Pros**

- _____
- _____

Cons

- _____
- _____

Scenario E: Heat-Flow Study in Granitic Terrane (Geothermal/Environmental)

Motivation/Goal. An energy agency is scoping geothermal potential and subsurface thermal risk for infrastructure. You must estimate **thermal conductivity**, $k(T)$ and **heat production** for granitic and metasedimentary units to forecast geotherms and transient responses. A number of pilot drill holes exist in the basin above, but do not intersect the granite. The only information about the granite comes from a gravity survey. How do you estimate the physical properties within the basin and granitic basement?

Your plan**2–3 Questions for the Geologist****Preferred method and why****Pros**

- _____
- _____

Cons

- _____
- _____

Scenario F: Geothermal Heat Flow at the Base of the Antarctic Ice Sheet (Geothermal/Glaciology)

Motivation/Goal. You are tasked with developing a geothermal heat flow model into the base of an ice sheet to support assessment of the ice sheet stability due to recent climate change. Several coastal nunataks provide limited outcrops with physical specimens and geochemistry, but their small areal extent and clustered distribution offer only modest constraints on subsurface lithology and properties beneath the ice interior. There is a relatively low-resolution seismic volume for the region and high-resolution gravity data. How would you develop the necessary **thermal conductivity** and **heat production** models?

Your plan**2–3 Questions for the Geologist****Preferred method and why****Pros**

- _____
- _____

Cons

- _____
- _____

Key Ideas

No single method suits every problem. Direct measurement on samples is most accurate but requires access; composition-based estimates are portable but depend on assumed mixing laws; geophysical inversion is non-unique but provides spatial coverage. The right choice depends on the data available, the scale of the model, and the required accuracy.

Geological questions constrain physical property models. Before selecting a method, a geophysicist must understand the geological context: What rock types are present? What are their pressure–temperature conditions? Is alteration or weathering likely? How well sampled is the region? The answers narrow the method options significantly.

Non-uniqueness is unavoidable. Different rock types or mixing geometries can produce similar effective properties. Uncertainty should be quantified and propagated into the final geophysical model, not treated as a source of error to be hidden.

Activity 2.C

Path Simulation

Purpose

To build intuition for how the geometric arrangement of rock components — not just their individual properties — determines the bulk effective property of a mixture.

Learning Outcomes

By the end of this activity, you should be able to:

- Predict which geometry (random, series, parallel) will produce the fastest or slowest transit, and connect this to the arithmetic, geometric, and harmonic mixing laws.
- Explain why the same two minerals can yield very different effective conductivities depending on their spatial arrangement.
- Read a travel-time or k_{eff} histogram and locate the harmonic, geometric, and arithmetic mixing-law predictions relative to the simulation results.
- Identify at least two ways a computer path simulator simplifies the real Earth, and assess whether those simplifications are likely to matter for a given geophysical application.

Background

When heat (or current, or fluid) moves through a mixture of two rock types A and B with conductivities k_A and k_B and volume fractions f_A and $f_B = 1 - f_A$, the effective conductivity k_{eff} depends on *how* the two phases are arranged:

Table 2: Mixing models for two component physical properties.

Arrangement	Geometry	Mixing law	k_{eff}
Layers $\parallel$ flow	parallel	Arithmetic (Voigt)	$f_A k_A + f_B k_B$
Layers $\perp$ flow	series	Harmonic (Reuss)	$(f_A / k_A + f_B / k_B)^{-1}$
Random granular	random	Geometric	$k_A^{f_A} k_B^{f_B}$

The arithmetic mean is always the *largest* (fastest paths bypass barriers), the harmonic mean the *smallest* (every path must cross every barrier), and the geometric mean sits in between with many similar random flow paths.

Travel time through a domain of width W is $t = W / k_{\text{eff}}$, so the geometry that maximises k_{eff} also minimises t .

Simulation

Your instructor will run the mixing-law path simulator on the projected screen. For each geometry listed in the table, note the simulation settings chosen, then record the mean travel time and effective conductivity reported by the simulation once ~ 100 trials have run. The dashed vertical lines on the histogram mark the harmonic (H), geometric (G), and arithmetic (A) mixing-law predictions.

Geometry	Simulation settings	Mean travel time	k_{eff} (sim)
Random granular			
Layered $\perp$ flow (series)			
Layered $\parallel$ flow (parallel)			
Random granular			
Layered $\perp$ flow (series)			
Layered $\parallel$ flow (parallel)			

Question 1. Ranking Geometry

Rank the three geometries from fastest (shortest travel time) to slowest. Which mixing-law prediction does each geometry match most closely?

Question 2. Comparing Mixing Laws

Where does the simulation's k_{eff} fall relative to the three dashed lines on the histogram?

Does it fall exactly on the predicted line, or does it scatter around it? Why might there be scatter?

Question 3. Effect of Contrast

How does increasing the contrast between k_A and k_B change the *spread* of the travel-time histogram?

How does it change the ratio $k_{\text{arith}}/k_{\text{harm}}$?

Question 4. Realism

In what ways does this simulator capture real rock behaviour?

In what ways does it over-simplify? List at least two assumptions the simulator makes that would not hold in a real geological setting.

Key Ideas

Geometry sets the mixing-law bound. The same two minerals can give k_{eff} values that span the full range from the harmonic to the arithmetic mean, depending solely on how they are arranged. This is the single most important idea in mixing-law theory: composition is necessary but not sufficient to predict a bulk property.

Travel time and effective conductivity are reciprocal. Because $t = W/k_{\text{eff}}$, the geometry that gives the highest k_{eff} (parallel) also gives the shortest travel time, and the geometry that gives the lowest k_{eff} (series) gives the longest. The same logic applies to electrical resistivity surveys, fluid flow in porous media, and seismic wavespeed in layered crust.

Simulation results scatter around analytic predictions. Each realisation of a random field is slightly different, so individual trial times vary. The analytic mixing laws predict the *expected* value of k_{eff} over many realisations; the histogram width reflects natural variability due to the finite, random arrangement of grains or layers.

Real rocks add complexity the simulator ignores. The simulator assumes isotropic, homogeneous components with sharp contacts, no fluid flow, no fractures, and steady-state conduction. Real rocks have anisotropic minerals, gradational contacts, multi-scale heterogeneity, and pressure- and temperature-dependent properties. Despite this, the mixing-law framework correctly identifies which geometries set the bounds on transport, which is why it remains in use across geophysics, materials science, and reservoir engineering.